



SMART ATTENDANCE SYSTEM USING FACE RECOGNITION

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Abstract

- The Smart Attendance System Using Face Recognition automates student attendance using a webcam. It recognizes registered students and records their attendance automatically in a MongoDB database, reducing manual work and errors.

Problem Statement & Objectives

- Problem Statement:
- Manual attendance is time-consuming, error-prone, and can allow proxy attendance.
- Objectives:
- Automate student attendance using face recognition.
- Identify registered students accurately.
- Store attendance records in MongoDB.
- Reduce manual work and attendance errors.
- Provide an easy-to-use attendance dashboard.

Existing vs Proposed System

Existing System	Proposed System
Manual attendance marking	Face recognition-based attendance
Time-consuming process	Fast and automated
More chances of errors	Reduces attendance errors
Paper/register-based records	MongoDB database
Proxy attendance possible	Student face verification
Manual attendance reports	Digital dashboard and records

Feasibility Study

- Technical: Easy to implement with Node.js and MongoDB.
- Economic: Low-cost solution.
- Operational: Simple and user-friendly.
- Time: Can be completed within the project duration.

Results & Testing

- Face recognition successfully identifies registered students.
- Attendance is automatically stored in MongoDB.

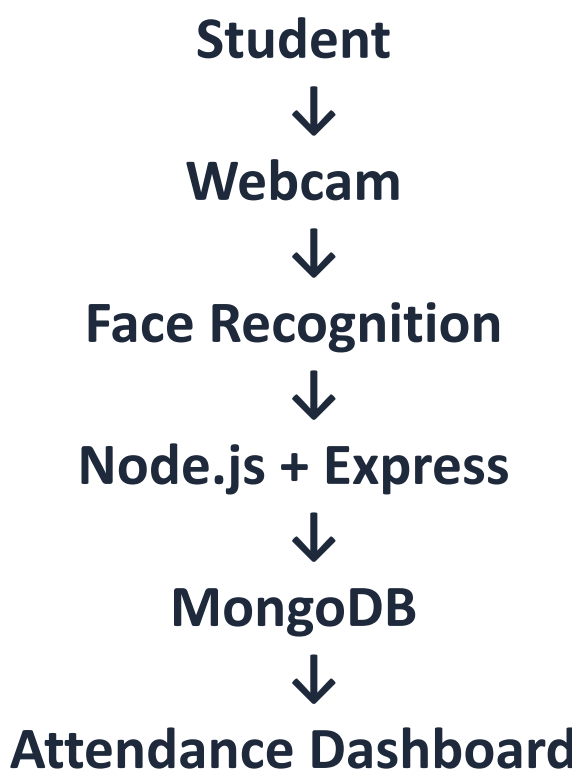
Conclusion

The Smart Attendance System Using Face Recognition provides a simple and automated way to record student attendance. It reduces manual effort, saves time, and stores attendance records securely in MongoDB.

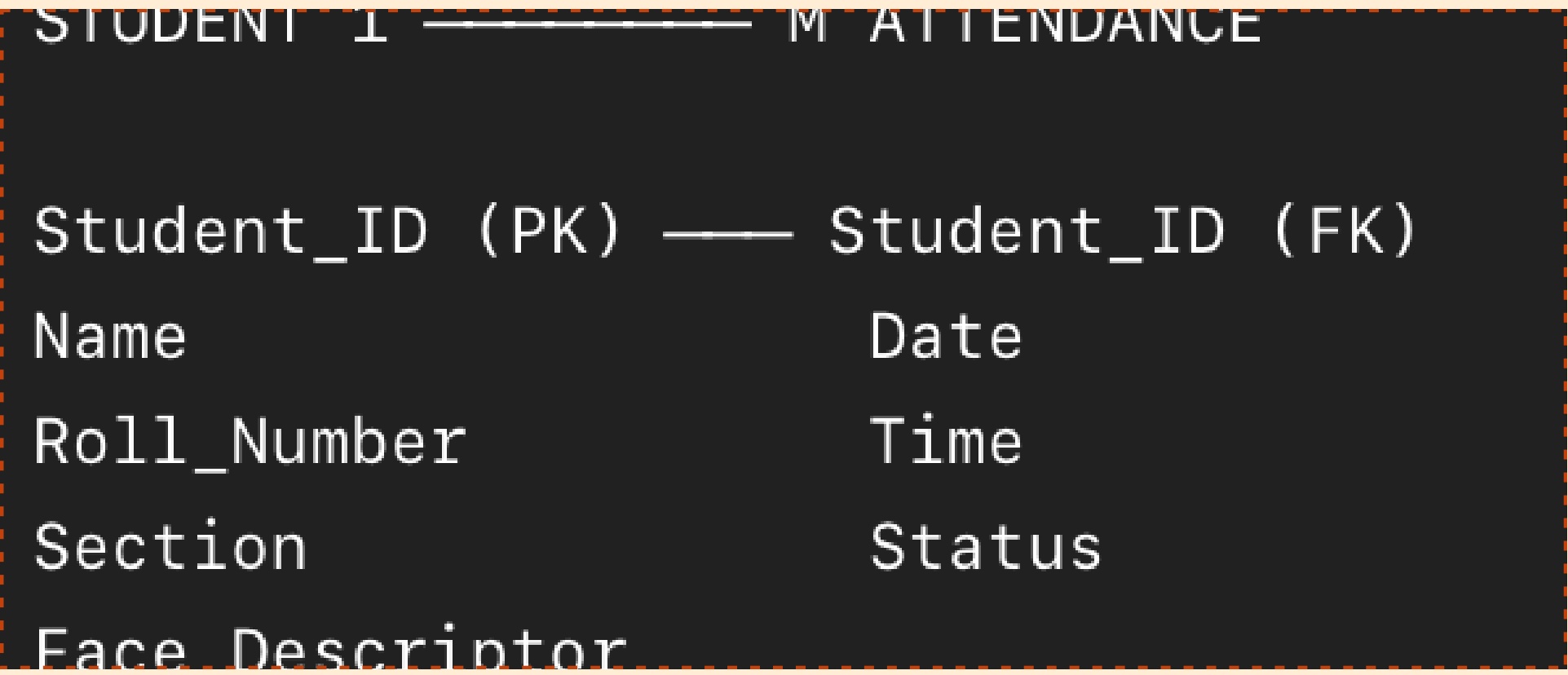
Tools & Technologies

- Frontend: HTML, CSS, JavaScript
- Backend: Node.js, Express.js
- Face Recognition: face-api.js
- Database: MongoDB
- Database Connectivity: Mongoose
- Development Tool: Visual Studio Code
- Browser: Google Chrome

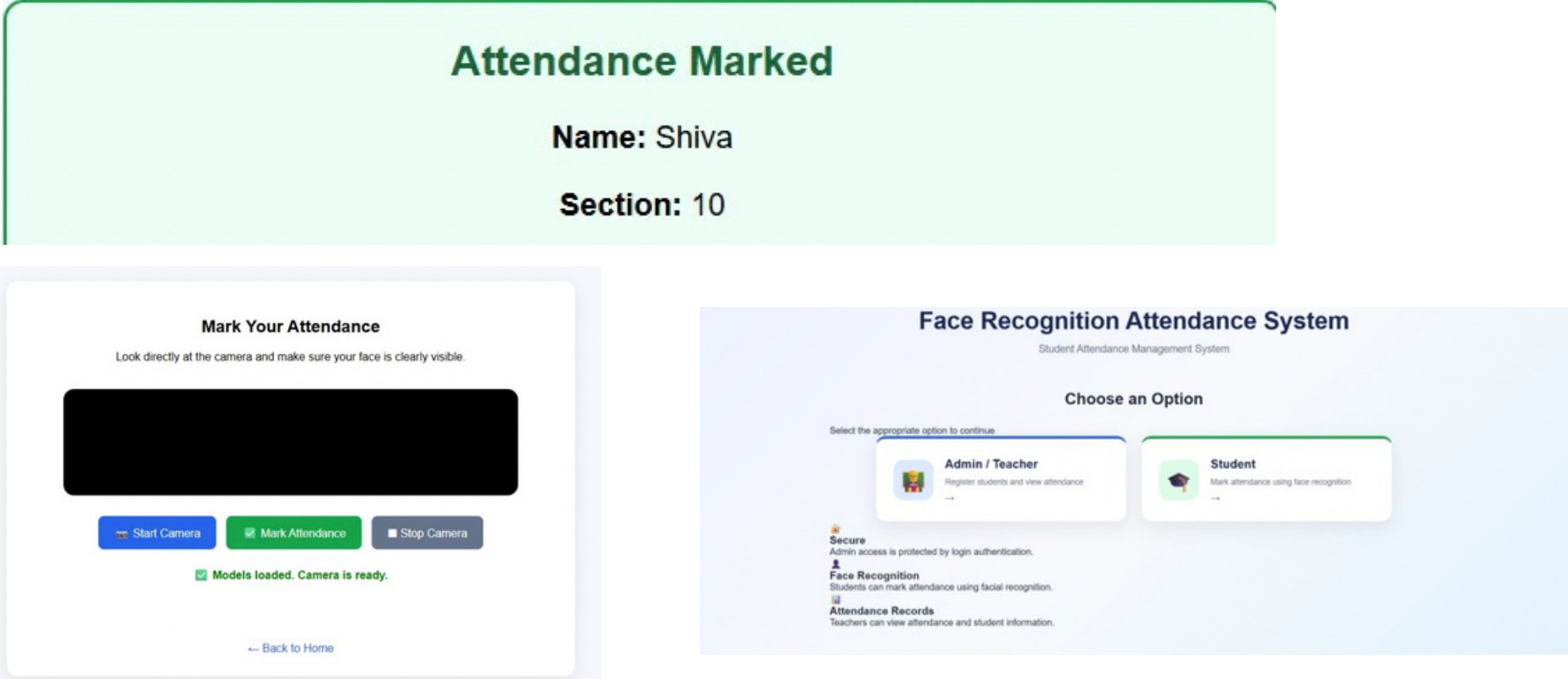
System Architecture



ER Diagram & Database Design



Implementation Screenshots



Future Scope

- Improve face recognition accuracy.
- Add mobile application support.
- Generate attendance reports automatically.
- Add multiple-admin support.